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HIGH-LEVEL DATA DESIGN DOCUMENT

Intelligent Deadline & Expiry Detection System

Project: proj11 | Lease: data_final_proj11_tt2884 | Branch: data/phase2-submission
Dataset: tanvitakavane/datanauts_project_cuad-deadline-ner-version2

System overview

This document describes the complete data architecture for the Intelligent Deadline & Expiry Detection feature added to Paperless-ngx. When a user uploads a document, the system extracts OCR text, identifies sentences containing dates, and classifies whether each date represents an actionable. High-confidence detections create automatic reminders; lower-confidence one's prompt user confirmation.

ML pipeline

Two fine-tuned models operate in sequence:

- dslim/bert-base-NER fine-tuned for date entity extraction (NER token classification)
- roberta-base fine-tuned for deadline classification (sequence classification)

Both models are trained on the cleaned CUAD dataset: tanvitakavane/datanauts_project_cuad-deadline-ner-version2 (Hugging Face). This version2 dataset uses only real CUAD contract data - no synthetic data is included. Synthetic data generation may be added in future iterations if the dataset size warrants it (current size < 5GB threshold allows for this option).

NER label schema

The canonical NER label set used by the training pipeline is: ["O", "B-EXP_DATE", "I-EXP_DATE", "B-START_DATE", "I-START_DATE", "B-DURATION", "I-DURATION"]. Downstream consumers such as the training pipeline may filter or remap event types based on their specific requirements. The canonical schema defined here remains the source of truth.

Classifier label schema

The classifier now supports 6 event types to match the full CUAD v2 dataset label distribution:

Label ID	Label Name	Description
0	none	Sentence contains no deadline or date-related information
1	expiration	Describes an expiration or termination clause (explicit date or duration-based)
2	effective	Describes the effective date or start date of the agreement
3	renewal	Describes an automatic renewal or extension clause
4	agreement	Describes the agreement date or contract execution date
5	notice_period	Describes a required notice period before termination or renewal

Note: Downstream consumers such as the training pipeline may filter or remap event types based on their specific requirements. The canonical schema defined here remains the source of truth.

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Data flow diagram

The following describes the end-to-end data flow from external source to model training:

```
EXTERNAL SOURCE
CUAD Dataset (HuggingFace: theatticusproject/cuad)
tanvitakavane/datanauts_project_cuad-deadline-ner-version2
        |
        v
  Ingestion Pipeline  (Docker, Chameleon VM)
  fetch --> clean --> convert parquet --> split --> upload
        |
   +-----+-----+
   |          |
   v         v
Chameleon     HuggingFace
Object Store  Dataset Repo
cuad-data-    version2
proj11-v2     (Git-backed, parquet format)
   |
   v
Batch Pipeline
(versioned splits)
versioned/v<TAG>/
manifest.json
   |
   +---> Training Team
        BERT NER fine-tuning
        train.jsonl / test.jsonl
   |
  v
Online Feature Service
FastAPI + Redis
(real-time inference path)
   |
   v
User feedback (confirm/edit/dismiss)
   |
   v
Monthly batch retraining --> new versioned dataset
```

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Data repositories

This section enumerates each data repository: what is stored, who writes it, and how it is versioned.

Chameleon object storage bucket (v2-active)

Attribute	Details
Bucket Name	cuad-data-proj11-v2
Site	CHI@TACC Swift - Publicly browsable
URL	https://chi.tacc.chameleoncloud.org/project/containers/container/cuad-data-proj11-v2
Previous Bucket	cuad-data-proj11 (v1 — retained for reference)

What is stored:

- raw/master_clauses.csv — Raw CUAD CSV (510 contracts, converted from parquet)
- data/cuad_cleaned.jsonl — Cleaned NER samples (510 records, JSONL format)
- data/train.jsonl — Training split (373 records)
- data/validation.jsonl — Validation split (68 records)
- data/test.jsonl — Test split (69 records, frozen)
- versioned/v<YYYYMMDD_HHMMSS>/ — Each batch pipeline run output
- versioned/v<TAG>/manifest.json — Pipeline manifest with source hash & split counts
- latest/ — Always pointing to most recent batch pipeline run
- eval_output/ — Evaluation and monitoring reports (EP1, EP2, EP3)

Who writes & when:
Written by the ingestion pipeline (Docker Compose on Chameleon VM) after each run. Also written by the batch pipeline (Q2.6) when it produces versioned training splits. The bucket is publicly readable so course staff can browse it at any time.

Versioning:
Every batch pipeline run creates a new versioned/ prefix. The manifest.json inside records: pipeline version tag, SHA256 hash of the source data, sample counts per split, split boundary logic, and a frozen_test: true flag. This allows any training run to be traced back to the exact data snapshot used.

Hugging face data repository

Attribute	Details
Repo	tanvitakavane/datanauts_project_cuad-deadline-ner-version2
Type	Public HuggingFace Dataset
Format	Parquet (auto-converted by HuggingFace from JSONL)
Access	load_dataset('tanvitakavane/datanauts_project_cuad-deadline-ner-version2')
Records	510 total: train=373, val=68, test=69

The HuggingFace dataset stores files in Parquet format (HuggingFace's standard). The ingestion pipeline downloads parquet files and converts them to JSONL for use in training and evaluation.

Redis in-memory feature store

Attribute	Details
Location	Docker container on Chameleon VM
Purpose	Online inference path - ephemeral feature cache
TTL	1 hour per document key

Pre-computed NER input features for each uploaded document, keyed by document_id. Each value is a JSON array of sentence-level feature objects: sentence text, token list, surrounding context, document type, section header, and a has_date_candidate flag. Not versioned — ephemeral by design. Redis is not used for training data.

PostgreSQL feedback database (planned)

User feedback events from Paperless-ngx: document_id, model prediction, user action (confirm/edit/dismiss/manual_add), corrected label, confidence score, created_at. Append-only. The monthly batch pipeline reads new records since its last run cutoff to produce updated training data.

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Data schema

Training sample schema (jsonl record)

Every record in train.jsonl, validation.jsonl, test.jsonl has the following fields (v2 dataset — converted from parquet):

Field	Type	Description
Filename	string	Source PDF filename (CUAD contract)
event_type	string	Deadline category: none, expiration, effective, renewal, agreement, notice_period (6 types)
ocr_text	string	Full OCR text of the contract document
agreement_date_iso	string	Agreement date in ISO 8601 format: YYYY-MM-DD (used for time-based split)
contract_date	string	Raw contract date (fallback for time-based split)
expiration_date_iso	string	Expiration date in ISO 8601 format
_version	string	Batch pipeline version tag, e.g. v20260419_182857 (batch runs only)
_split	string	Split assignment: train, val, or test (batch runs only)
_source_hash	string	SHA256 prefix of source file — links output to exact input (batch only)

Note on downstream consumers: The training pipeline uses NER labels ["O", "B-EXP_DATE", "I-EXP_DATE", "B-START_DATE", "I-START_DATE", "B-DURATION", "I-DURATION"] and classifier labels {0:none, 1:expiration, 2:effective, 3:renewal, 4:agreement, 5:notice_period}. Downstream consumers such as the training pipeline may filter or remap event types based on their specific requirements. The canonical schema defined here remains the source of truth.

Candidate selection & data leakage prevention

How training samples were selected

Dataset v2 (Current):
tanvitakavane/datanauts_project_cuad-deadline-ner-version2 contains cleaned CUAD data only (no synthetic data). The dataset has 510 contracts with 6 event types: none, expiration, effective, renewal, agreement, notice_period. Files are stored in Parquet format on HuggingFace and converted to JSONL by the ingestion pipeline.

Candidate Selection Criteria:

- ocr_text length >= 15 characters (filters empty/corrupted records)
- Filename must be present (ensures traceability to source contract)

- 13 records with empty ocr_text excluded (2.5% of dataset — within acceptable threshold)

Leakage prevention

- Time-based split: Contracts split by agreement_date_iso - pre-2005 to train, 2005-2010 to validation, 2011+ to test. Never split randomly.
- Contract-level grouping: All clauses from one contract go to the same split. No clause from a post-2010 contract appears in train.
- Frozen test set: Test split is written once and never used for retraining. Recorded as frozen_test: true in manifest.json.
- Source hash tracking: SHA256 of cuad_cleaned.jsonl recorded in every manifest — links any training run to exact data snapshot.

Shared inference – json samples

Clf_sample.json: [Datanauts-Intelligent-Deadline-Expiry-Detection/samples/clf_sample.json at training/phase2-submission · Bhanuu01/Datanauts-Intelligent-Deadline-Expiry-Detection]

Ner_sample.json: [Datanauts-Intelligent-Deadline-Expiry-Detection/samples/ner_sample.json at training/phase2-submission · Bhanuu01/Datanauts-Intelligent-Deadline-Expiry-Detection]

Data generator event types:

The data generator (Q2.4) now generates events matching all 6 event types from the v2 dataset:

Event Type	Example Sentence Template
expiration	This Agreement shall expire on {d}, unless terminated earlier.
effective	This Agreement is effective as of {d} upon execution by both parties.
renewal	The Agreement shall automatically renew for successive one-year terms.
agreement	This Agreement is entered into as of {d}, by and between the parties.
notice_period	Either party may terminate upon sixty (60) days prior written notice.
none	Each party agrees to keep confidential all proprietary information.

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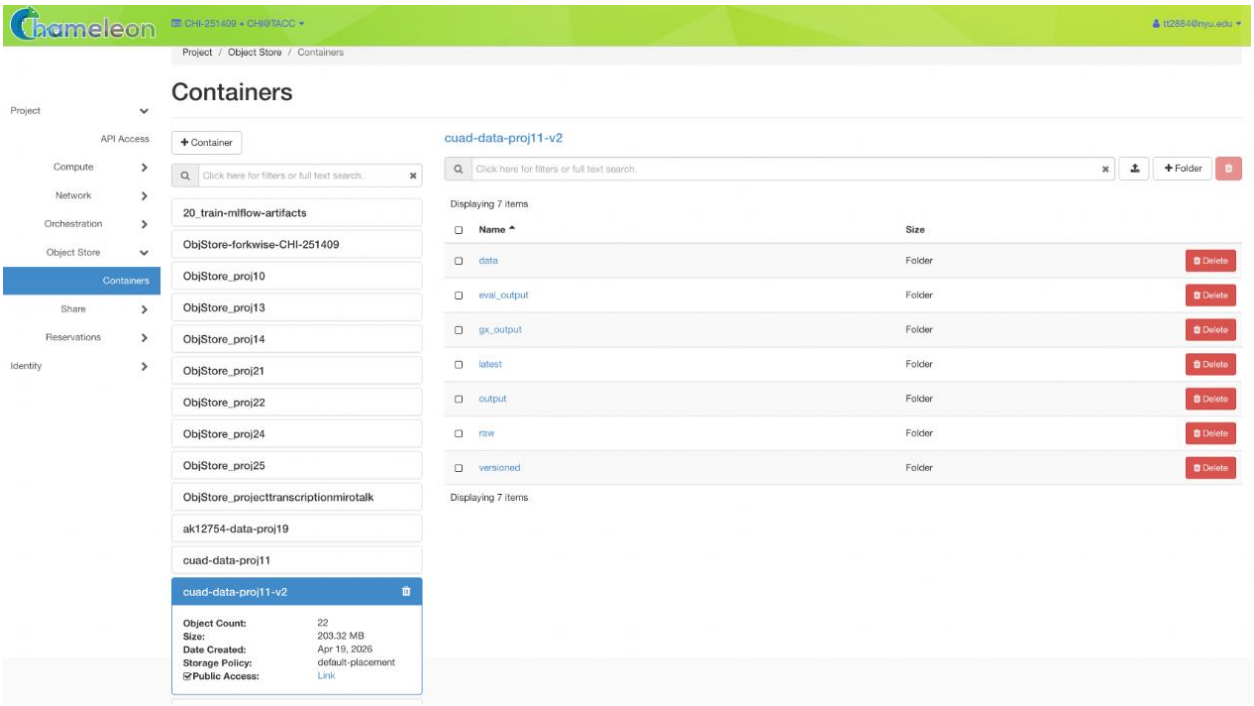
Repository structure

All artifacts are committed to branch: Datanauts-Intelligent-Deadline-Expiry-Detection

Path in Repo	Contents
pipeline/	Q2.3 Ingestion pipeline (Dockerfile, docker-compose.yml, scripts/)
data_generator/	Q2.4 Data generator — 6 event types (Dockerfile, docker-compose.yml, generator.py)
online_features/	Q2.5 Feature service — FastAPI + Redis (Dockerfile, docker-compose.yml, feature_service.py)
batch_pipeline/	Q2.6 Batch pipeline — versioned datasets (Dockerfile, docker-compose.yml, batch_pipeline.py)
evaluation_monitoring/	Final implementation — EP1 ingestion check, EP2 training quality, EP3 drift monitor
object_storage/	Q2.2 Object storage README — cuad-data-proj11-v2 bucket documentation

New container containing all files for re-implementation of initial submission (Apr 6) and final submission (Apr 20):